UNIVERSIDAD NACIONAL DE COLOMBIA

Respuestas - Taller de Álgebra Lineal

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Respuestas

Sistemas de ecuaciones y forma matricial

1.

$$A = \begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix}, \quad b = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$$

$$[A|b] = \left[\begin{array}{cc|c} 2 & 3 & 7 \\ 1 & -1 & 1 \end{array}\right]$$

2.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad b = \begin{pmatrix} 6 \\ 3 \\ 4 \end{pmatrix}$$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{array}\right]$$

Eliminación de Gauss

1.

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 1 \end{array}\right] \xrightarrow{F_2 - 3F_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -14 \end{array}\right]$$

$$-7y = -14 \Rightarrow y = 2$$

$$x + 2(2) = 5 \Rightarrow x = 1$$

$$\boxed{(x, y) = (1, 2)}$$

2.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 3 & 14 \\ 3 & 2 & -1 & 2 \end{array}\right]$$

Realizando operaciones elementales:

$$\left.\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & -1 & -4 & -16 \end{array}\right.$$

$$\left.\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & 0 & -\frac{11}{3} & -\frac{46}{3} \end{array}\right.$$

$$z = 4, \quad y = 2, \quad x = 0$$

$$\boxed{(x, y, z) = (0, 2, 4)}$$

Determinantes

1.

$$\det(A) = 2(7) - 3(5) = 14 - 15 = -1$$

$$\boxed{\det(A) = -1}$$

2.

$$\det(B) = 1(4) - 2(3) = 4 - 6 = -2$$

$$\boxed{\det(B) = -2}$$

3.

$$\det(C) = 3\begin{vmatrix} 0 & 1 \\ 4 & 3 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} + 2\begin{vmatrix} 2 & 0 \\ 1 & 4 \end{vmatrix}$$

$$= 3(0 \cdot 3 - 1 \cdot 4) - 1(2 \cdot 3 - 1 \cdot 1) + 2(2 \cdot 4 - 0 \cdot 1)$$

$$= 3(-4) - 1(5) + 2(8) = -12 - 5 + 16 = -1$$

$$\boxed{\det(C) = -1}$$